

SAI KOUSHIK

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PROFESSIONAL SUMMARY

Full Stack Engineer with **4 years** of production experience building payment and financial platforms, full-stack web applications, and AI-powered automation across **TypeScript, React, Python, and Java**. Designed and shipped REST APIs, third-party integrations, and accessible React frontends on **AWS**, scaling services to **10K+** daily transactions at **sub-300ms** p95 latency, maintaining reliable settlement across **500K+** monthly transactions and **70M+**-record datasets, and cutting p95 latency **28%**. Currently shipping ChangeGuard AI, a multi-agent platform reducing deployment risk review from **15 minutes** to under **30 seconds**, with strong grounding in auth, security, testing, and CI/CD across regulated, fintech-adjacent workflows.

PROFESSIONAL EXPERIENCE

Chicago Public Schools

Chicago, IL

Research AI Engineer

Aug **2025** – Present

Stack: Python, FastAPI, TypeScript, Anthropic Claude, LangChain, Langfuse, Pydantic, pytest, Docker, GitHub API, GitHub Actions

- Architected ChangeGuard AI, a FastAPI multi-agent system processing **100+** real GitHub pull requests, cutting deployment risk assessment from **15 minutes** to under **30 seconds** by orchestrating agents across code churn, IaC changes, blast radius, and incident signal analysis, directly targeting Change Failure Rate, a core DORA metric.
- Designed and maintained a third-party GitHub integration: a production webhook service handling **300+** events with HMAC-SHA256 security and constant-time validation, returning structured risk assessments under **200ms** via FastAPI async tasks, Pydantic schema validation, pytest evaluation suites, and Docker deployment.
- Built an LLM orchestration pipeline coordinating **5** specialized agents into structured risk scores, PR comments, rollout recommendations, and escalation gates, achieving **85%+** ground-truth agreement across a **25-case** evaluation suite with full Langfuse tracing and quality monitoring.

Thomson Reuters

Hyderabad, India

Full Stack Engineer

Aug **2021** – Jul **2024**

Key Client: FedEx (logistics payment platform, 2+ year enterprise engagement)

Stack: Java, Spring Boot, React, TypeScript, AWS ECS/Fargate/Lambda/RDS/S3, Redis, Kafka, Terraform, Docker, MS SQL, Okta

- Delivered full-stack payment features for FedEx – a responsive React SPA frontend and event-driven Java/Spring Boot services on AWS ECS Fargate – supporting **10K+** daily transactions across **4** microservices with **sub-300ms** p95 latency; integrated Okta SSO via SAML 2.0 for **10K** enterprise users and visualized **120M+** transaction records, accelerating reconciliation workflows by **40%**.
- Designed clean service, domain, and persistence boundaries and cut p95 latency **28%** (**420ms** → **302ms**) via Completable-Future async processing, HikariCP pool tuning (**50** → **150**), and batch operations reducing DB round-trips by **55%** and customer timeouts by **25%**.
- Scaled relational query performance **2.3x** on a **70M**-record dataset (**2.8s** → **1.2s**) through range partitioning, composite indexes, and optimistic locking; architected multi-AZ AWS infrastructure with Terraform, ECS Fargate autoscaling, and RDS Multi-AZ reducing provisioning from **4 hours** to **18 minutes**.
- Built CI/CD pipelines with **85%** test coverage (JUnit, Mockito, Pact, Testcontainers) and blue-green deployments cutting pipeline time **40%** with **sub-6-minute** rollbacks; partnered directly with FedEx product, design, and QA stakeholders to shape requirements and SLOs, reducing rework by **30%** while mentoring **2** junior engineers.

PROJECTS

Cruddur: Ephemeral Micro-Blogging Platform

[GitHub](#)

Feb **2025** – Jun **2025**

Tech: React, Flask, Docker, AWS ECS Fargate, PostgreSQL, DynamoDB, Momento, Cognito, CloudFormation, Lambda, X-Ray, Honeycomb

- Deployed a React SPA and Flask backend on AWS ECS Fargate behind CloudFront with Cognito auth; provisioned CloudFormation infrastructure with PostgreSQL, DynamoDB, and Momento caching reducing query latency **58%** under **\$12/month**, with X-Ray and Honeycomb observability and Lambda processing in under **4** seconds.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL

Frontend: React, Responsive & Accessible UI, Tailwind CSS, Bootstrap

Backend & APIs: FastAPI, Flask, Spring Boot, REST APIs, Microservices, Event-Driven Architecture, Webhooks, Kafka

AI & LLM: Anthropic Claude, LangChain, LLM Orchestration, Multi-Agent Systems, RAG, Langfuse, Prompt Engineering

Cloud & DevOps: AWS (ECS, Fargate, Lambda, RDS, S3, IAM, CloudWatch), GitHub Actions, Jenkins, Terraform, Docker, Kubernetes

Databases & Auth: PostgreSQL, MS SQL, DynamoDB, Redis, OAuth2, OIDC, SAML 2.0, JWT, Okta, Cognito, RBAC

Testing & Observability: pytest, JUnit, Mockito, Testcontainers, Blue-Green Deployments, AWS X-Ray, Langfuse, SLOs/SLIs

Certifications: AWS Certified Developer Associate, AWS Certified Cloud Practitioner